Colimeter Development Plan

1. Determine LED requirements
    a) Type: use CREE J2835 series, only narrow band colors.
        - All are available from Digikey at about $0.20 to $0.62.
        - Violet, Royal Blue, Blue, Cyan, Green, Amber, Red, Photo Red.
        - Avoid other colors are wide band or multi-band or outside of known camera sensitivity response.
        - Include all 8 valid colors in calibration unit, up to 4 in field unit
2. Determine LED geometry
    a) Test of 4 corner, recessed plane design (To Be Confirmed)
    b) Geometries: "Cmm": "C" = 4 corner geometry, mm is distance from light to target in mm.
        i) Get image in 1 corner
        ii) Process as if sum of 4 corners on 5 strips in horizontal and 7 strips in vertical.
        iii) Evaluate for evenness of illumination.
3. Determine camera geometry
    a) Focal length
4. Determine measurement and calibration methods
    a) Lighting at sensor peaks, sensor slopes, or both?
    b) Analysis scheme
    c) Test strip calibration scheme
    d) Reference tile calibration scheme
    e) Scheme for translating test strip calibration to new reference tiles.
    f) Scheme for sharing calibrations and tiles.
5. Determine LED power requirements and Resistors
    - Simplified for field unit
    - More complex for calibration unit
6. Determine Camera: Module 2 or 3?
    - Maybe both? Would require building 2 calibration machines, and running every sample through both.
7. Design battery power supply
8. Design and make PCBs
    a) PCBs for LEDs (LEDs and Resistors)
        - Field machine and Calibration machine patterns?

b) PCB for mobo
    - Rpi, lighting controls, interface, buttons, connectors, power supply.
    - Field machine and Calibration machine patterns?

c) Assemble all parts and verify.

9. Make Calibration tiles

a) Base tiles: correct clay, leather hard, pre-scored on backs

b) Mason stains and slip base: mix standard slips

c) Don't forget white and gray.

d) Needed tiles:
    - Calibration machine color sets
    - Field machine color sets

e) Labeling and identification: can we mark the backs?

10. Design 3D printed housing

a) Install software

b) Body – size, joints, lining,

c) Base – joints, sample tray, reference chips

d) Camera, lighting, and mobo bulkhead

e) Top and interface mounting

f) Verify

11. Programming

a) Camera

b) Lighting

c) Physical interface

d) Web interface

e) Timer for test kit reagents

f) Sample analysis

g) Recalibration analysis

h) Archiving results and metadata

i) Documentation